

FTP Server

vsftpd · Linux System & Network Administration

1. Service Fundamentals

FTP (File Transfer Protocol) is a TCP client-server protocol used to upload, download and manage files over a network. It uses a control connection (port 21) and separate data connections. Plain FTP does not encrypt credentials or data — prefer SFTP or FTPS for sensitive transfers.

Item	Details
Service / Package	FTP · vsftpd
Control Port	TCP 21
Data Ports	Active: TCP 20 · Passive: high ports (configurable)
Main Config	/etc/vsftpd.conf
Service Unit	vsftpd.service
Encryption	None (plain FTP) — use SFTP/FTPS for security

2. Active vs Passive Mode

Mode	Who opens data connection	Note
Active	Server → Client	Often blocked by client firewalls / NAT
Passive	Client → Server	Preferred; open a limited passive port range

3. Installation

Update package list and install vsftpd:

Command:

```
sudo apt update
```

Command:

```
sudo apt install vsftpd
```

Enable and start the service:

Command:

```
sudo systemctl enable --now vsftpd
```

Create a backup of the default configuration:

Command:

```
sudo cp /etc/vsftpd.conf /etc/vsftpd.conf.bak
```

4. Configuration

Open the main configuration file:

Command:

```
sudo nano /etc/vsftpd.conf
```

Set the following essential options:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
chroot_local_user=YES
allow_writeable_chroot=YES
listen=YES
listen_ipv6=NO
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40100
```

anonymous_enable=NO — disable guest login
local_enable=YES — allow local Linux users
write_enable=YES — allow uploads
chroot_local_user=YES — jail users inside their home directory

pasv_min/max_port — restrict passive data ports for easier firewall rules

Apply the configuration:

Command:

```
sudo systemctl restart vsftpd
```

5. Verification

Check service status:

Command:

```
sudo systemctl status vsftpd
```

Confirm TCP 21 is listening:

Command:

```
sudo ss -lntp | grep :21
```

Test from a client:

Command:

```
ftp username@SERVER_IP
```

Inside the FTP session you can use: *ls · pwd · cd · get · put · bye*

6. Configuration Reference

Directive	Purpose
anonymous_enable=NO	Disable anonymous login
local_enable=YES	Allow local Linux users
write_enable=YES	Permit uploads and file changes
chroot_local_user=YES	Jail users to their home directory
allow_writeable_chroot=YES	Required when home is writable under chroot
pasv_min_port / pasv_max_port	Limit passive data port range

7. Troubleshooting

Problem	Check	Solution
Cannot login	local_enable, password	Enable local users; verify account
Login OK, transfer fails	Passive mode, firewall	Use Passive; open ports 40000-40100
Permission denied on upload	write_enable, directory perms	write_enable=YES; fix ownership
500 OOPS error	journalctl -u vsftpd	Set allow_writeable_chroot=YES

8. Security

- Disable anonymous access.
- Use chroot_local_user=YES so users cannot escape their home.
- Restrict the passive port range and open it only on the firewall.
- Do not expose plain FTP to the public Internet for sensitive data — use SFTP instead.

9. Real-World Scenario

A company needs an internal file-drop server. The administrator installs vsftpd, disables anonymous login, enables local users with chroot, opens TCP 21 and ports 40000-40100 on the internal firewall, and creates dedicated Linux accounts for each department.

10. Essential Commands

Command	Purpose
sudo apt install vsftpd	Install FTP server
sudo systemctl restart vsftpd	Apply configuration
sudo systemctl status vsftpd	Check service status
sudo ss -lntp grep :21	Verify control port
sudo journalctl -u vsftpd -f	Live logs
ftp user@host	Client connection test

11. Important Points (Interview / Viva)

1. FTP uses TCP 21 for control and a separate connection for data.
2. Active = server initiates data connection; Passive = client initiates (preferred).
3. Package: vsftpd · Config: /etc/vsftpd.conf
4. write_enable=YES allows uploads; chroot jails users to home.
5. Plain FTP is unencrypted — prefer SFTP for confidential data.